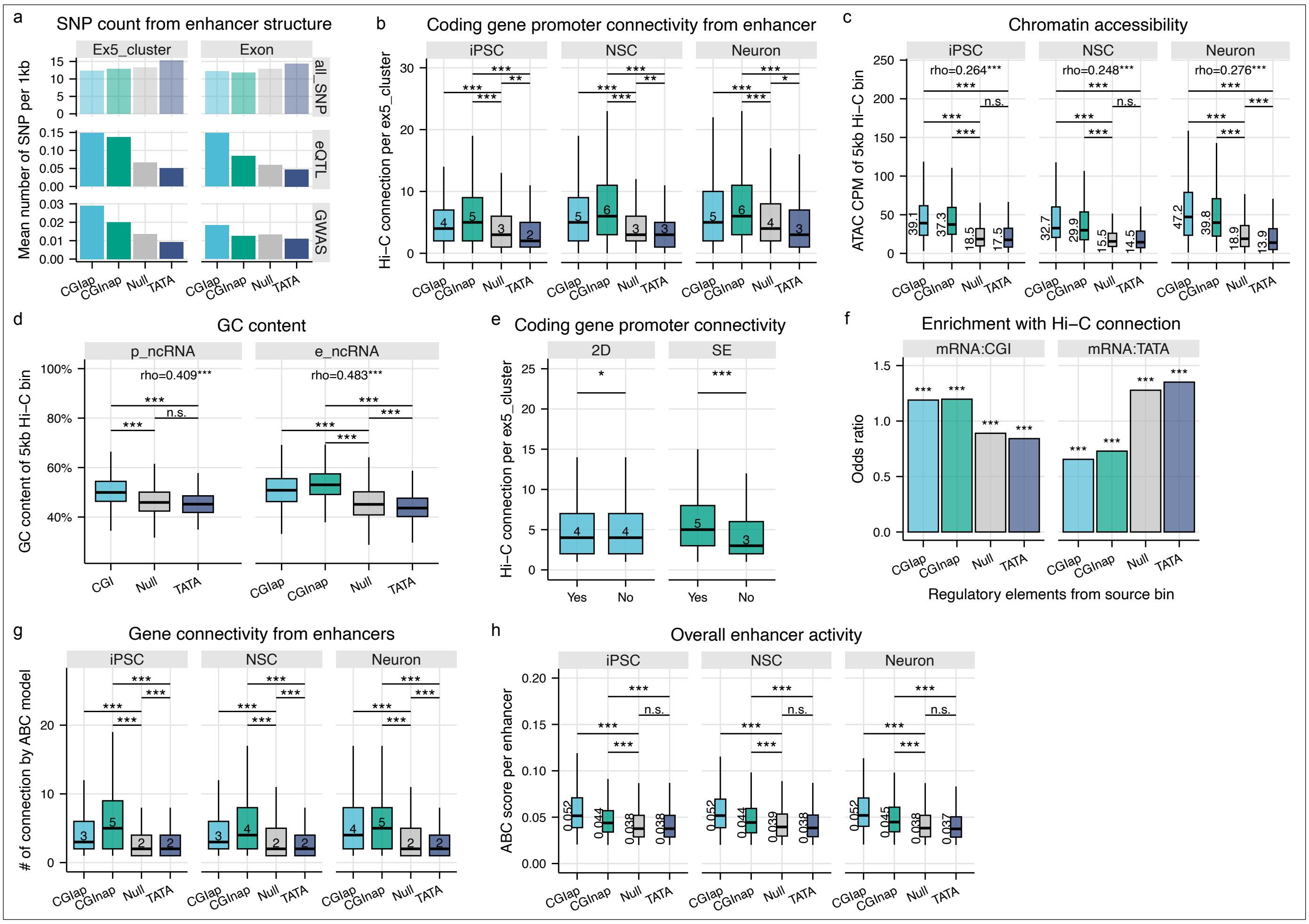

Extended Data Figure 8 | Genetic variants and enhancer connectivity.

a, Number of SNPs associated with e_ncRNAs grouped by mutually exclusive regulatory elements of ex5_clusters. SNP counts were normalized by the length of the ex5_cluster or transcript model (per kb). The "all_SNP" catalog refers to common SNPs from the NIH dataset. **b**, Hi-C interactions (5 kb resolution, FDR < 0.01, ≥5 reads) were used to infer ncRNA–mRNA connectivity via ex5_clusters. Only e_ncRNA ex5_clusters with ≥1 connection were included. Bins containing ex5_clusters with multiple regulatory elements (CGI and TATA box) were excluded. Median connection counts per ex5_cluster are shown, analyzed separately for each of the three cell types. **c**, Chromatin accessibility from matching ATAC-seq samples was mapped to 5 kb bins containing a single regulatory element (multi-element bins excluded). Median read counts and Spearman correlation between chromatin accessibility and Hi-C connectivity (as in panel **b**) are shown. **d**, GC content was calculated for 5 kb bins as in (**c**), and Spearman correlation with Hi-C connectivity (as in **Figure 6d**) is shown. **e**, Hi-C connectivity across all three cell types was compared against tCRE directionality and super-enhancer (SE) clustering. **f**, Two-sided Fisher's exact test was used to assess enrichment across regulatory element combinations. ncRNA source was grouped as in panel **b**; mRNA promoters were classified by presence or absence of CGI or TATA box. Odds ratios are shown for each combination. **g**, Gene connectivity of enhancers estimated by ABC-model. Analysis was performed on merged tCREs and linked to the features of ex5_clusters. Merged tCREs containing ex5_clusters with multiple regulatory elements (CGI and TATA box) were excluded. Only the enhancers with at least one connection with ABC score > 0.02 were included. **h**, Same analysis as (**g**) showing the mean ABC score for each enhancer. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.